

John Peterson

Software Architect | Engineering Leader | Cloud-Native & Distributed Systems

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Software Architect and Engineering Leader with 10+ years of experience designing and delivering cloud-native SaaS platforms, distributed systems, and event-driven architectures that process millions of daily events and support 5,000+ employees and tens of thousands of customer portal users. Proven track record building greenfield platforms from the ground up, leading distributed engineering organizations, and establishing the architectural standards and engineering culture that drive lasting technical excellence. Passionate about solving hard systems problems and mentoring engineers into the technical leaders that scale organizations.

KEY ACHIEVEMENTS

- Architected a greenfield SaaS platform supporting 5,000+ employees and tens of thousands of customer portal users, replacing a third-party CRM and field operations suite across a 30-person engineering organization.
 - Built an event-driven integration platform processing millions of events per day across 10+ distributed services, forming the backbone of a mission-critical cloud-native architecture.
 - Reduced deployment cycles from days to minutes and enabled hundreds of production deployments per day through CI/CD transformation and modern engineering practices.
 - Raised automated test coverage from 0% to approximately 98% across multiple engineering teams, significantly improving release confidence and long-term maintainability.
 - Built an IT organization from a single team member to approximately twenty, supporting a \$40M company through a period of rapid growth and digital transformation.
 - Led the migration of more than 500,000 customer records between CRM platforms while maintaining data integrity and business continuity throughout.
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CORE COMPETENCIES

Software Architecture · Distributed Systems · Event-Driven Architecture · Cloud-Native Applications · System Design · Microservices · Identity & Access Management · Observability & Monitoring · Platform Engineering · CI/CD & DevOps · Technical Leadership · Engineering Management · Team Building & Mentorship · Performance & Scalability · API Design & Integration · Domain-Driven Design (DDD) · Agile Delivery

TECHNICAL SKILLS

Languages

Go · Python · PHP · JavaScript · SQL · Java

Cloud & Infrastructure

AWS · Kubernetes (EKS) · ECS · Docker · Docker Swarm · Pulumi · ArgoCD · Traefik · LocalStack

Data & Messaging

PostgreSQL · MySQL · Redis · RabbitMQ · InfluxDB

Observability

OpenTelemetry · Grafana · Datadog · Prometheus

Identity & Security

Okta · SCIM · OAuth2 · OpenID Connect (OIDC) · SSO

Engineering Practices

CI/CD · GitOps · Automated Testing · Agile/Scrum · Technical Leadership · Mentorship · Team Building

PROFESSIONAL EXPERIENCE

Apptive Environmental | Software Architect

June 2024 – Present

Utah, United States

- Architect for Aspyr, a greenfield cloud-native SaaS platform built by a 30-person engineering organization to replace a third-party CRM, sales, field operations, and routing suite supporting 5,000+ employees and tens of thousands of customer portal users.

- Delivered foundational platform services powering Aspyr — including identity and authorization, centralized address validation and verification, short-link services, event-driven integrations, and PDF document generation APIs.
- Built an event-driven integration platform backed by RabbitMQ that serves as the backbone of the Aspyr architecture, processing millions of events per day and enabling loosely coupled communication across 10+ distributed services.
- Established centralized authentication and authorization services, including custom SCIM-compliant provisioning endpoints and Okta SSO integrations supporting enterprise identity lifecycle management across platform applications.
- Engineered cloud infrastructure utilizing AWS EKS, ECS, Pulumi, and ArgoCD, enabling highly scalable and repeatable production environments with GitOps-based deployment workflows.
- Drove organization-wide observability by implementing OpenTelemetry-based distributed tracing, metrics, and logging with collector infrastructure that provides end-to-end visibility across all distributed services.
- Designed autoscaling strategies utilizing KEDA, Prometheus, and AWS CloudWatch metrics to efficiently scale asynchronous workloads under highly variable demand.
- Collaborated with fellow architects to define service boundaries, architectural constraints, and long-term technical direction across a distributed, service-oriented platform.
- Mentored engineers and technical leaders through design reviews, technical pairing sessions, and internal technical writing — elevating architectural thinking across the organization.
- Introduced CI/CD and modern engineering practices that reduced deployment cycles from days to minutes, enabling hundreds of production deployments per day.

Aptive Environmental | Software Engineering Manager

November 2021 – June 2024

Logan, Utah

- Led four fully remote software development teams of 8–16 onshore and offshore engineers, accountable for the technical health, quality, and delivery of 4–6 business-critical services.
- Owned the full engineering management lifecycle — hiring, onboarding, performance management, promotions, coaching, and difficult personnel decisions — while building high-performing and accountable teams.
- Developed and promoted three engineers into Technical Lead positions through deliberate mentorship, career planning, and leadership development.
- Established engineering standards and processes including API documentation, branching strategies, code review practices, and technical quality expectations that elevated consistency and developer productivity across teams.
- Introduced CI/CD pipelines and automated testing that reduced developer feedback cycles from days to minutes, transitioning teams from biweekly releases to multiple production deployments per day.
- Led engineering transformation that raised automated test coverage from 0% to approximately 98%, significantly improving release confidence and long-term maintainability.
- Drove organization-wide observability initiatives utilizing distributed tracing, metrics, and logging that enabled teams to proactively detect and resolve incidents before customers were impacted.
- Implemented product instrumentation and technical KPIs using InfluxDB and Grafana, giving teams the visibility to measure feature adoption, system performance, and business outcomes.
- Introduced caching strategies and performance optimization initiatives that improved application responsiveness and scalability across multiple services.
- Owned capacity planning, staffing decisions, sprint planning, roadmap execution, and vendor evaluations to ensure teams could effectively deliver strategic initiatives.
- Selected as one of 13 employees out of approximately 2,500 company-wide for Aptive's High Potential executive leadership development program.

Aptive Environmental | Software Engineer

September 2021 – November 2021

United States

- Designed, built, and maintained RESTful APIs and backend services within a cloud-native microservices architecture.
- Delivered business capabilities and platform enhancements within Agile Scrum teams, with strong emphasis on code quality and automated testing.
- Developed performant, scalable, and maintainable software solutions aligned with long-term architectural objectives.
- Participated in architecture discussions and technical design sessions to ensure solutions met business requirements and scalability goals.

Fox Pest Control | Software Architect

July 2020 – September 2021

Logan, Utah

- Designed and led implementation of a custom payroll platform supporting 300+ door-to-door sales representatives, automating highly complex commission calculations and multi-tier compensation workflows. The system remains mission-critical and in active use today.
- Designed and led implementation of a custom task management and operational tracking platform utilized by 80+ office staff to manage day-to-day branch operations, track work execution, and provide real-time visibility into business KPIs. The platform remains actively used across the organization.
- Built and led a software development team delivering custom internal web applications and backend services supporting operations across multiple business departments.
- Designed communication patterns and integration flows between backend services, third-party SaaS platforms, and internal databases, enabling reliable data synchronization and business process automation.
- Partnered closely with business stakeholders to gather requirements, define project scope, prioritize initiatives, and translate complex operational needs into technical solutions.
- Established Agile Scrum development processes with Jira, Git, and Bitbucket, improving delivery predictability, project visibility, and team collaboration.
- Simplified development environment setup and cross-team collaboration through Docker-based workflows and standardized local environments.
- Introduced CI/CD practices via Bitbucket webhooks and Jenkins pipelines, significantly reducing deployment complexity and improving release consistency.
- Designed software architectures grounded in OOP, SOLID principles, DRY principles, and clean architecture to maximize maintainability and long-term scalability.
- Implemented asynchronous job processing, caching strategies, and memory optimization techniques to improve application performance and responsiveness.
- Raised automated unit test coverage from 0% to approximately 70% across web applications.
- Mentored developers through architecture reviews, design discussions, pair programming, and active removal of technical roadblocks.
- Contributed directly to web applications and backend services in PHP, Laravel, Lumen, JavaScript, HTML, CSS, jQuery, and MySQL.
- Established engineering standards and development practices that improved code quality, developer productivity, and long-term maintainability.

Fox Pest Control | Director of Information Technology & Software Development

March 2017 – July 2020

Logan, Utah

- Built the IT organization from the ground up, growing the department from a single team member to approximately twenty — supporting a \$40M company through a period of rapid expansion.
- Directed software developers in designing, building, and maintaining custom business applications in Python, JavaScript, PHP, MySQL, HTML, and CSS.
- Established software development workflows and source control practices with Git, improving team collaboration, release management, and code quality.
- Designed, hosted, and maintained on-premises MySQL databases supporting business-critical applications and operational systems.
- Architected and maintained a centralized MySQL data warehouse supporting company-wide reporting, analytics, and business intelligence.
- Oversaw the successful migration of more than 500,000 customer records between CRM platforms, ensuring data integrity and business continuity throughout.
- Built Python-based data integration pipelines that consumed third-party APIs and synchronized operational data into internal systems and reporting platforms.
- Led a team of data analysts in developing mission-critical business intelligence dashboards and KPIs in Microsoft Power BI.
- Designed data flows between third-party platforms, APIs, and the centralized data warehouse, enabling executive reporting and data-driven decision making.
- Performed deep data validation and investigative analysis to ensure reporting accuracy and resolve complex data quality issues.
- Designed, hosted, and maintained on-premises web servers, databases, and infrastructure supporting business-critical applications.
- Deployed Ubuntu virtual machines hosting network services including SNMP monitoring, Syslog aggregation, WiFi controllers, internal web applications, and external SFTP services.
- Led IT support teams providing technical assistance to 200+ sales representatives and 70 office staff.
- Oversaw network infrastructure, software deployments, VOIP systems, hardware lifecycle management, and enterprise technology initiatives across a rapidly growing organization.

- Continuously researched and introduced new technologies and operational processes that improved organizational efficiency, scalability, and business capability.

Adilas, LLC | Full Stack Software Developer (Part-Time)

March 2020 – August 2020

- Maintained and enhanced a large legacy web application built on Adobe ColdFusion, MySQL, HTML, CSS, and JavaScript.
- Delivered a modernized homepage experience while preserving full compatibility with existing backend architecture and business workflows.
- Improved application performance by implementing session-based caching and eliminating unnecessary system configuration lookups.
- Collaborated with stakeholders to deliver incremental improvements while balancing modernization goals with legacy system constraints.

Bridgerland Technical College | Information Technology Instructor (Part-Time)

August 2018 – January 2019

- Developed hands-on curriculum and lab exercises covering networking, systems administration, and core information technology concepts.
- Mentored students in software development and IT principles, practices, and problem-solving methodologies.
- Prepared students for technology careers by grounding instruction in practical skills and industry best practices.
- Made complex technical concepts accessible and cultivated lasting enthusiasm for technology and lifelong learning.

C E R T I F I C A T I O N S

- AWS Certified Solutions Architect – Associate
- AWS Certified Cloud Practitioner
- ITIL Foundations
- CompTIA Security+
- CompTIA Network+
- CompTIA A+
- CompTIA Project+
- Software Design and Architecture Specialization
- Object-Oriented Design
- Design Patterns
- Service-Oriented Architecture
- TechWorld with Nana – DevOps Practitioner Certification

E D U C A T I O N**Bachelor of Science, Software Development**

Western Governors University

Web Development Certification

Bridgerland Technical College

Information Technology Certification

Bridgerland Technical College

Computer Science Studies

Utah State University

Associate of Science, Business

Snow College

H O N O R S & L E A D E R S H I P**High Potential Program**

Aptive Environmental

- Selected as one of 13 employees out of approximately 2,500 company-wide for executive leadership development.

Web Development Advisor

Bridgerland Technical College

